

Rising Skill, Falling Skill: Why a Single Baseline Cannot Tell You Whether an Ocean Forecast Model Has Learned Anything

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Abstract. Statistical forecasts of ocean state are conventionally scored against *persistence*, the last observed value. We show this single baseline is not merely incomplete but actively misleading, because it fails in a direction that flatters the model. Persistence decays with forecast horizon while a seasonal cycle does not, so a model that has learned nothing but the time of year exhibits skill against persistence that *increases* with horizon — the signature normally read as evidence of a strong model. We audit 7 operational ocean variables across 24 globally distributed sites and four horizons (1 d to 30 d), using one gradient-boosted framework with no variable-specific code, and score every model against both persistence and a leakage-controlled day-of-year climatology. Median skill against persistence rises from 0.230 at 1 d to 0.379 at 30 d; over the same range median skill against climatology falls from 0.580 to -0.055. The two baselines therefore support opposite conclusions about the same models, and 9 model-horizon combinations beat persistence while losing to climatology outright. We further test whether these globally pooled models transfer to locations they never saw, using leave-one-site-out validation blocked in space and time simultaneously (672 held-out fits). Transfer is real but costly: median held-out skill at 7 d is 0.109 against 0.151 pooled, with 65.5% of held-out sites retaining positive skill. We report 5 model-horizon combinations that lose to persistence, and argue that reporting both baselines across multiple horizons should be a minimum standard for this class of work.

Keywords: Ocean forecasting · Forecast verification · Skill scores · Baseline selection · Spatial cross-validation · Gradient boosting

1 Introduction

An ocean forecast is useful only relative to the cheapest thing one could have done instead. In operational oceanography that reference is almost always *persistence*: predict that tomorrow equals today. Persistence is a famously hard baseline on autocorrelated geophysical series, and a model that beats it is normally taken to have learned something.

This paper argues that persistence alone cannot support that inference, and that the way it fails is systematically misleading rather than merely noisy.

The argument is structural. Persistence error grows with the forecast horizon, because the ocean decorrelates from its present state. The error of a *seasonal climatology* does not: knowing that it is February is exactly as informative thirty days ahead as it is one day ahead. A skill score computed against persistence therefore has a denominator that inflates with horizon while the numerator of a purely seasonal model does not. The consequence is that a model which has learned nothing beyond the annual cycle will display skill against persistence that *increases* with horizon — precisely the pattern a reader interprets as a model with genuine long-range predictive content.

We encountered this pattern empirically. In a framework trained across 7 ocean variables, skill against persistence rose with horizon for 6 of them. Whether that reflected real forecast skill or a seasonal artefact was, with persistence as the only baseline, unanswerable. This paper resolves it by adding the baseline that discriminates, and finds that both readings are present in the same set of models — including 9 model–horizon combinations that beat persistence while losing to climatology outright.

1.1 Contributions

1. A **two-baseline skill audit** over 7 operational ocean variables $\times$ 4 horizons $\times$ 24 global sites (28 models), scoring every model against both persistence and a leakage-controlled day-of-year climatology. The two baselines order the same models differently, and the disagreement is a systematic function of horizon rather than noise.
2. A **diagnosis of the horizon–skill inversion**, separating genuine long-range skill from a seasonal artefact, and identifying the variables for which a climatology — cheaper to compute and to serve — is the rational operational choice.
3. **Leave-one-site-out spatial validation** blocked in space and time simultaneously (672 held-out fits), testing the assumption that licenses serving a globally pooled model at arbitrary coordinates.
4. **Negative results reported in full**, including 5 model–horizon combinations that lose to persistence outright. These identify a class of high-inertia variables for which the approach is contraindicated.
5. A **practical decision procedure** (Fig. 6) mapping the pair $(S_{\text{pers}}, S_{\text{clim}})$ to a deployment recommendation.

2 Related Work

Skill scores relative to a reference forecast are standard in numerical weather prediction, and are conventionally reported against climatology, persistence, or both [1,4]. The concern raised here is not that climatology is unknown as a baseline — it is long-established in NWP — but that in the recent machine-learning-for-ocean literature persistence is frequently the sole reference, and the

specific interaction between baseline choice and forecast horizon is rarely made explicit.

Data-driven ocean and atmospheric forecasting has moved rapidly from convolutional recurrent architectures for sea surface temperature [7] to global neural weather models [5,3]. Evaluation practice has not kept pace uniformly: gains are often reported for a single variable at a single horizon against a single baseline. Our contribution is orthogonal to architecture. We deliberately hold the model class fixed — gradient-boosted trees, one configuration, no variable-specific code — so that variation in the results is attributable to the variable, the horizon and the baseline rather than to model tuning.

The leakage discipline we adopt for the climatology follows standard practice in ecological and environmental modelling, where fitting any component of a baseline or label definition on the evaluation period is a known and consequential error [6]. Our spatial validation follows the same literature’s argument against random k -fold splits on spatially autocorrelated data.

3 System Architecture

Figure 1 shows the framework. It is deliberately uniform: one acquisition path, one feature builder and one model class serve every variable, and nothing in the pipeline branches on which variable it is training. Adding a variable is a configuration entry, not a code change. This uniformity is what makes a cross-variable audit meaningful — differences in the results cannot be attributed to differences in the implementation.

4 Data and Methods

4.1 Variables and sites

We study 7 ocean variables spanning physics, surface waves and biogeochemistry (Table 1), all drawn from operational products: Copernicus Marine Service analysis and forecast fields for the physical and biogeochemical variables, and the same provider’s wave products for the wave variables. Variables were included on the criterion that a complete 24-site history was available in our archive at the snapshot date — a data-availability criterion applied before any model was scored, discussed further in Section 7.

Each variable is sampled at 24 sites chosen to span the major ocean basins and dynamical regimes: western boundary currents (Gulf Stream, Kuroshio, Agulhas), eastern boundary upwelling systems (Peru, Benguela, Canary), subtropical gyres, marginal and shelf seas, and the Southern Ocean. Site locations are shown in Fig. 5. The record spans 2024-06-11 to 2026-08-09 at daily resolution.

4.2 Forecast framework

For a target variable y at horizon h , features at time t comprise lagged values of y and its covariates at lags $\{1, 3, 7, 14, 30\}$ days; rolling mean, standard deviation,

Table 1. The 7 variables audited. “Feature rows” is the pooled count across all 24 sites after feature construction and target alignment.

Variable	Unit	Sites	Feature rows
bottom temperature	degC	24	18,960
maximum wave height	m	24	18,888
mean wave period	s	24	18,888
significant wave height	m	24	18,888
water salinity	PSU	24	18,960
water temperature	degC	24	18,960
wave direction	deg	24	18,888

minimum and maximum over windows of $\{7, 14, 30\}$ days; first differences; linear trend coefficients over $\{7, 30\}$ days; calendar and cyclical encodings of day of year; and static site descriptors (latitude, longitude, ocean depth, basin). All rolling windows close at t inclusive, which is safe because $h \geq 1$ is enforced. Exactly one forward shift exists in the pipeline, in target construction.

Models are LightGBM regressors [2] with fixed hyperparameters across all variables and horizons (400 trees, learning rate 0.05, 31 leaves). No per-variable tuning was performed: tuning would confound the cross-variable comparison that is the point of the audit.

Delta targets. Models regress on the *change* $y(t+h) - y(t)$ rather than the level $y(t+h)$. A regression tree partitions feature space into piecewise-constant regions and therefore cannot represent the identity function $y(t+h) = y(t)$ — which is exactly what persistence is. Fitted on levels, the framework scored worse than persistence at every horizon tested. Under a delta target persistence becomes the constant zero function, which a tree represents exactly, and the same data and features yield positive skill. Predictions are decoded by adding the model output to the anchor $y(t)$. For circular variables (wave direction) the target is the signed angular change wrapped to $(-180^\circ, 180^\circ]$ and decoding is modulo 360° .

Globally pooled models. One model per (variable, horizon) is fitted across all 24 sites jointly, with site coordinates and depth as features, rather than one model per site. This is what allows the framework to score arbitrary coordinates rather than only the sites it was fitted on — and it is the assumption Section 5.4 tests directly.

4.3 Baselines

Persistence. $\hat{y}_{\text{pers}}(t+h) = y(t)$, the last observed value at feature time. This is also the anchor of the delta target, so a delta model that outputs zero *is* persistence exactly.

Climatology. For each site s and day of year d , the climatology is the mean of all training observations at that site whose day of year falls within ± 15 days of d , treating the year as circular so that 31 December neighbours 1 January:

$$\hat{y}_{\text{clim}}(s, d) = \frac{1}{|W(d)|} \sum_{t \in W(d)} y_s(t), \quad W(d) = \{t : \delta(\text{doy}(t), d) \leq 15\}, \quad (1)$$

where δ is circular day-of-year distance. The window is necessary rather than cosmetic: the record is roughly 2.2 years, so a bare (site, day-of-year) mean would average two samples and be mostly noise — a strawman the model would beat for the wrong reason. A ± 15 -day window pools approximately 60 samples per estimate. For circular variables the mean is the circular mean, computed via atan2 of the mean sine and cosine; the arithmetic mean of 359° and 1° is 180° , which would make the baseline point in exactly the wrong direction.

Leakage control. The climatology is refitted *inside every cross-validation fold* on that fold’s training rows only (shaded in Fig. 2). Fitting a climatology once over the full record would leak the evaluation period into the baseline’s own definition, flattering the baseline and understating the model — an error in the “safe” direction, and therefore one that is easy to leave in place. Fit and application are implemented as two separate functions so that fitting on the wrong rows requires deliberate effort, and the discipline is enforced by a unit test that shifts the second year of a synthetic series by a known constant and asserts the baseline’s error on that period recovers the shift.

4.4 Validation

Figure 2 shows both experiments end to end.

Rolling-origin cross-validation. Five expanding-window folds, ordered in time, with an embargo gap equal to the forecast horizon between each training window and its test window — a training row within h days of the test window shares its target period. Random k -fold splitting is never used. The pooled multi-site frame is sorted by time so that a fold boundary separates *dates*, and no site’s future can leak into another site’s past.

Skill score. For a reference forecast r ,

$$S_r = 1 - \frac{\text{RMSE}_{\text{model}}^2}{\text{RMSE}_r^2}, \quad (2)$$

so $S_r > 0$ means the model beats reference r , $S_r = 0$ is parity, and $S_r < 0$ means the reference is better. We report S_{pers} and S_{clim} for every model. Where the climatology cannot answer for some test rows (a day of year no site in the training fold observed), both model and baseline are scored on the intersection, and the reduced count is reported rather than the rows being silently dropped.

Table 2. Skill against persistence (S_p) and against climatology (S_c) at each horizon. Bold marks a negative score — the baseline winning. Reading across a row shows the divergence; reading down the S_c columns shows it is systematic.

Variable	$h = 1$		$h = 3$		$h = 7$		$h = 30$	
	S_p	S_c	S_p	S_c	S_p	S_c	S_p	S_c
bottom temperature	0.253	0.992	-0.033	0.935	-0.032	0.823	-0.109	0.502
maximum wave height	0.249	0.572	0.327	0.047	0.362	-0.042	0.386	-0.100
mean wave period	0.230	0.561	0.310	0.043	0.374	-0.026	0.429	-0.055
significant wave height	0.272	0.580	0.331	0.045	0.360	-0.044	0.379	-0.108
water salinity	0.050	0.958	-0.168	0.753	-0.089	0.480	0.100	-0.023
water temperature	0.075	0.958	0.009	0.817	0.139	0.633	0.430	0.187
wave direction	0.090	0.445	0.112	0.029	0.151	-0.157	0.163	-0.254

4.5 Reproducibility

All experiments run from a fixed archival snapshot (2026-08-10), pinned so that window boundaries — and therefore every cached provider request — are identical across reruns. Without this pinning each rerun re-requests the record from upstream providers and, because those providers intermittently return partial responses, occasionally scores a slightly different dataset, making two runs of the same experiment non-comparable. Every figure and table cell in this paper is generated programmatically from the experiment output; no number is transcribed by hand.

5 Results

5.1 The two baselines disagree, and disagree with horizon

Figure 3 is the central result. For each variable it plots skill against persistence and skill against climatology as a function of horizon.

The two curves systematically diverge. Median skill against persistence rises from 0.230 at 1 d to 0.379 at 30 d, while median skill against climatology falls from 0.580 to -0.055 over the same range. This is the structural prediction of Section 1 realised in data: persistence becomes easy to beat as the horizon grows because the ocean decorrelates from its present state, while climatology becomes *harder* to beat because the seasonal cycle is an increasing share of what remains predictable.

The practical consequence is that the two baselines license opposite statements about the same models. Reported against persistence alone, this framework appears to improve with lead time. Reported against climatology alone, it appears to degrade toward the seasonal cycle. Both are true; neither is complete. Per-variable skill scores are given in Table 2, and the full error decomposition in the appendix table.

5.2 What survives the seasonal cycle

The inversion does not mean every model is purely seasonal. At 30 d, skill against climatology remains positive for most variables, meaning they carry information the seasonal cycle does not. But 9 model–horizon combinations have $S_{\text{clim}} < 0$ while $S_{\text{pers}} > 0$: these models beat the last observed value and lose to the seasonal cycle. Under the reporting convention this literature most often uses, they would be presented as successes.

The correct reading is that long-horizon skill measured against persistence is substantially — and for some variables entirely — seasonal, and only the climatology comparison separates the two components. Variables whose S_{clim} approaches or crosses zero at long range are those for which the model has effectively reduced to a seasonal lookup, and for which a climatology would be the rational operational choice.

5.3 Where the framework fails

5 model–horizon combinations have $S_{\text{pers}} < 0$: the model is worse than doing nothing. These concentrate in the high-inertia subsurface variables, and the pattern is physically coherent. For a variable whose autocorrelation time greatly exceeds the forecast horizon, persistence is close to optimal and the residual the delta model is asked to predict is close to pure noise; fitting it adds variance without adding signal. This is a property of the variable rather than a defect of the implementation, and it is useful precisely because it is predictable in advance from the autocorrelation structure — it identifies where this class of model should not be deployed at all.

Notably these are not the same variables that fail against climatology. Some subsurface variables lose to persistence while beating climatology by a wide margin; some wave variables do the reverse. The two baselines fail in different places, which is the strongest available argument for reporting both.

5.4 Spatial transfer: leave-one-site-out

Serving a globally pooled model at arbitrary coordinates rests on an assumption that is rarely tested: that the model generalises to ocean it never saw during training, rather than interpolating among its training sites. We test this directly. For each variable, horizon and site, we refit on the remaining 24 – 1 sites and score on the held-out one (672 fits).

The split is blocked in space *and* time. Training uses the earlier 80% of the record at the 23 retained sites; testing uses the later 20% at the held-out site. A purely spatial hold-out would leave the model trained on the same dates it is tested on, measuring interpolation across a shared period rather than transfer. The climatology baseline is held to the same standard: it is fitted only on the 23 training sites, so for the held-out site it falls back to the nearest fitted site by great-circle distance — the spatial analogue of what the model itself is being asked to do.

Table 3. Leave-one-site-out results. “Med.” is the median skill against persistence across the 24 held-out sites; n^+ counts sites retaining positive skill. Bold marks a negative median — a variable that does not transfer.

	$h = 1$		$h = 3$		$h = 7$		$h = 30$	
Variable	Med.	n^+	Med.	n^+	Med.	n^+	Med.	n^+
bottom temperature	0.23	16	-0.17	10	-0.45	3	-3.02	4
maximum wave height	0.21	23	0.23	23	0.32	22	0.40	23
mean wave period	0.17	23	0.26	21	0.36	22	0.41	23
significant wave height	0.21	23	0.21	21	0.27	21	0.36	23
water salinity	0.14	16	-0.03	10	-0.17	6	-0.04	10
water temperature	0.13	18	0.06	14	0.11	16	0.52	20
wave direction	0.12	18	0.12	16	0.15	20	0.24	20

Transfer is real but costs skill. At 7 d, median skill at held-out sites is 0.109 against 0.151 pooled, and 65.5% of held-out site-fits retain positive skill. That the majority remain positive supports serving the pooled model at unseen coordinates; that the median falls means the pooled figure overstates what a user clicking on open ocean should expect, and a deployed system should say so. Per-variable summaries are in Table 3.

Figure 5 shows transfer is not spatially uniform. Sites in dynamically energetic regions — western boundary currents and upwelling systems — are systematically harder to predict from elsewhere, the expected result: their variability is driven by mesoscale processes with short decorrelation scales that a model fitted on distant sites cannot resolve.

6 Discussion

6.1 A decision procedure

Figure 6 states the practical consequence of these results as a procedure. The pair $(S_{\text{pers}}, S_{\text{clim}})$, evaluated across more than one horizon and checked against a held-out site, maps onto a small number of clearly distinguishable recommendations — three of which are “do not deploy this model” for reasons that a single baseline cannot surface.

6.2 Implications

Report both baselines. Persistence and climatology are individually cheap and jointly diagnostic, and their disagreement carries information neither provides alone. A model that beats persistence at long range while losing to climatology has not learned to forecast; it has learned what month it is. That failure mode is invisible under a single-baseline convention, and it fails in the flattering direction.

Baseline choice interacts with horizon, so single-horizon evaluation hides it. The divergence in Fig. 3 is only visible across horizons. A paper reporting one horizon — however carefully — can report a number that is correct and still leave the reader unable to tell which of two very different models produced it.

Architecture is not the lever these results point to. We held the model class fixed deliberately. Nothing here argues that gradient-boosted trees are the right architecture; the results argue that *evaluation* is currently the binding constraint on interpreting such comparisons. A more expressive architecture evaluated against persistence alone at a single horizon would produce a number that is neither comparable to these nor obviously more informative.

Pooling across sites is defensible but should be reported as such. The leave-one-site-out results support serving a globally pooled model at arbitrary coordinates, with the caveat that pooled metrics overstate performance at unseen locations, and that the overstatement is largest exactly in the energetic regions users are most likely to care about.

7 Limitations

Record length. The archive spans roughly 2.2 years. This is the most consequential limitation and it bears directly on the central comparison: a climatology estimated from two annual cycles is weakly constrained, and the ± 15 -day window is what makes it a serious baseline rather than a noisy one. A longer record would strengthen the climatology and, we expect, *reduce* the measured S_{clim} — our headline comparison is therefore conservative in the direction that favours the model, and the qualitative conclusion is unlikely to reverse. It also means we cannot separate the seasonal cycle from interannual variability.

Variable coverage. The 7 variables reported are those with complete 24-site coverage in the archive at the snapshot date. Others were excluded on data availability, not on their results. The criterion was applied before any model was scored, so it does not select on outcome, but the set is not a random sample of ocean variables — it is skewed toward well-observed operational products.

Reanalysis as ground truth. The “observations” are operational analysis products, not in-situ measurements. They are themselves model output constrained by assimilated observations, so our scores measure agreement with an analysis rather than with the ocean. This is standard for work at this spatial coverage, but it bounds what the results can claim, particularly in sparsely observed regions where the analysis is least constrained.

Circular variables are modelled linearly. Wave direction is decoded circularly and scored with circular error, but the underlying regression treats the wrapped delta as a linear target: a veer of 5° across north trains as -355° . The principled fix — predicting vector components and deriving the angle — is not applied here, and the direction results should be read with that caveat.

Fixed hyperparameters. No per-variable tuning was performed. This is necessary for the cross-variable comparison but means every individual model is likely below its own achievable ceiling. The audit measures the framework as configured, not the best attainable skill per variable.

8 Conclusion

Skill against persistence and skill against climatology are not interchangeable summaries of the same quantity, and the difference between them is not noise: it is a systematic function of forecast horizon that can invert the qualitative conclusion. Across 7 ocean variables, 24 sites and four horizons, median skill against persistence rises with lead time while median skill against climatology falls, and 9 model-horizon combinations beat persistence while losing to the seasonal cycle outright.

We also find that globally pooled models do transfer to sites they never saw, with 65.5% of held-out sites retaining positive skill at 7 d, but at a real cost that pooled metrics conceal, concentrated in dynamically energetic regions.

The recommendation is cheap to adopt: report both baselines, across more than one horizon, and report the cases where the baseline wins.

Reproducibility. Experiments run from a pinned archival snapshot (2026-08-10). The climatology implementation, its leakage tests, the experiment driver and the asset generator that produces every figure and table are included in the accompanying source tree (`forecasting/climatology.py`, `tests/test_climatology.py`, `scripts/run_paper_experiments.py`, `scripts/build_paper_assets.py`).

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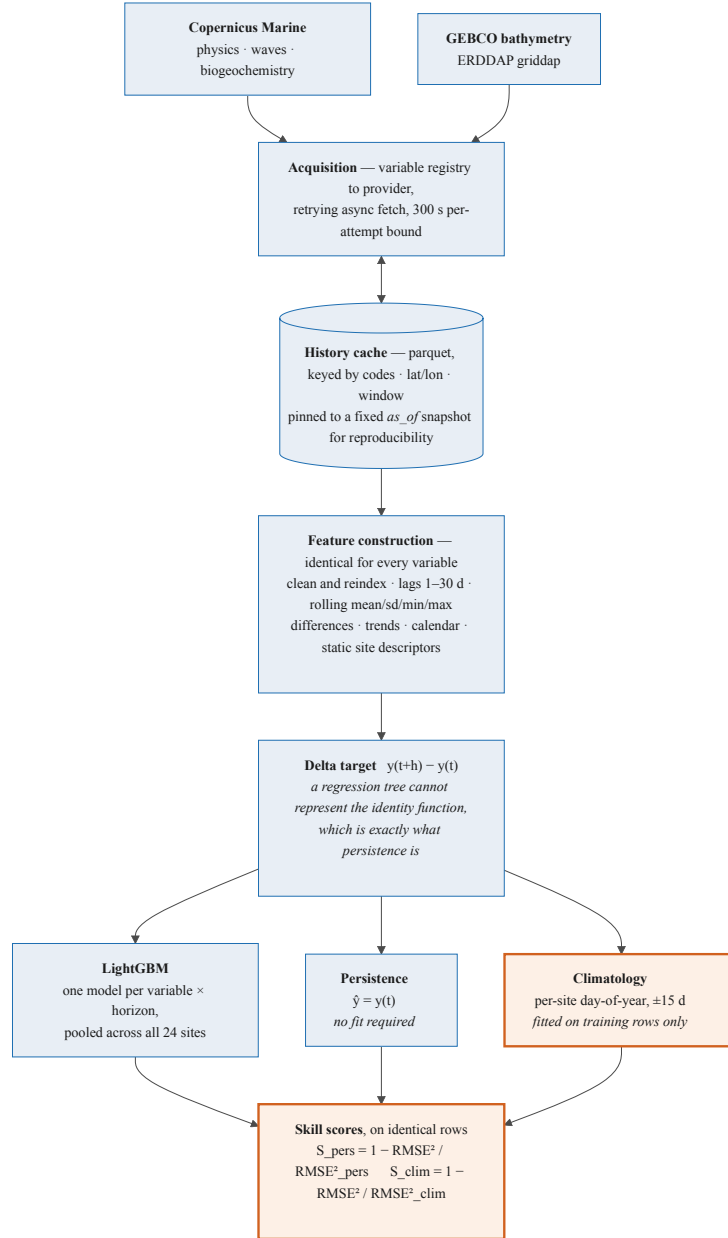


Fig. 1. Framework architecture. One acquisition path, one feature builder and one model class serve all 7 variables. The three predictors at the bottom are scored on identical rows. The contribution of this paper is shaded: the climatology baseline and the second skill score it makes possible.

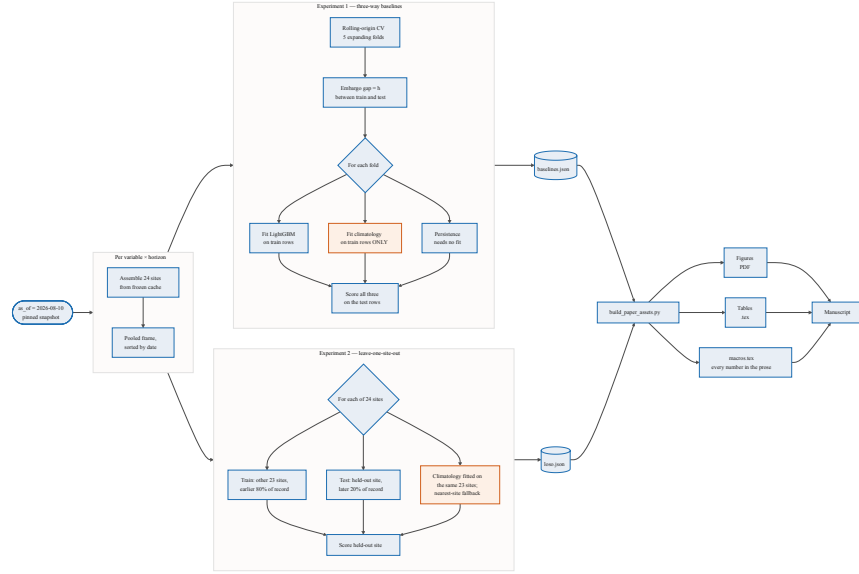


Fig. 2. Experiment data flow. A pinned `as_of` snapshot makes every provider request cache-stable, so the experiments are reproducible and run offline. Shaded boxes are the two places a baseline is fitted, and both are fitted strictly inside the fold they are scored in.

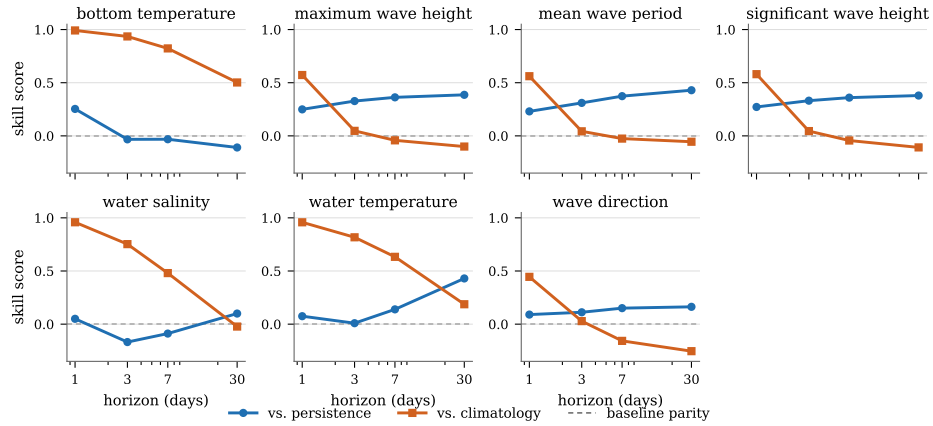


Fig. 3. Skill against two baselines, by forecast horizon, for each of the 7 variables. Skill against persistence (blue circles) and against climatology (orange squares) move in *opposite* directions with horizon for most variables. The dashed line is parity; points below it mean the baseline wins. A reader given only the blue curve would conclude these models improve with lead time.

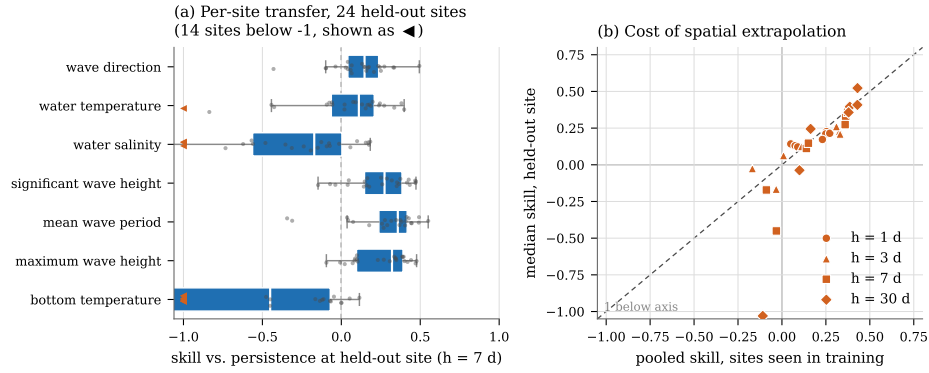


Fig. 4. Leave-one-site-out transfer. (a) Skill at each held-out site (7d); each dot is one site scored by a model that never saw it. (b) Pooled skill at sites seen in training against median skill at held-out sites; the dashed line is equality, and points below it are the cost of spatial extrapolation.

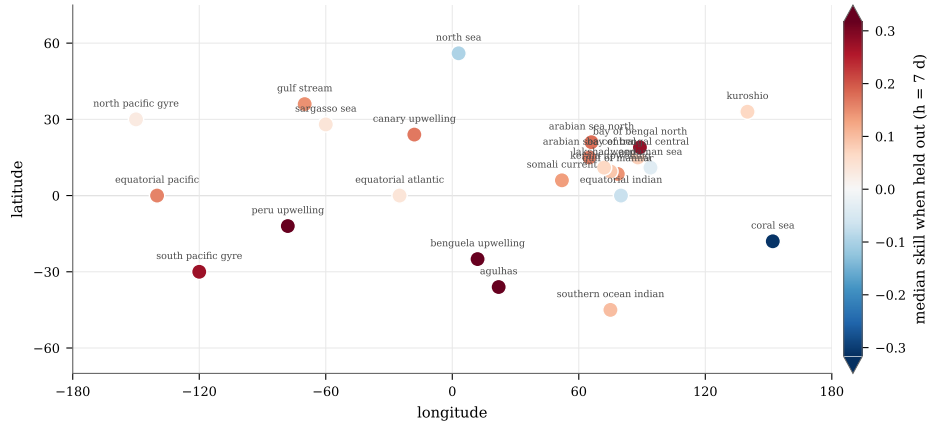


Fig. 5. The 24 evaluation sites, coloured by median skill across variables when that site is held out entirely (7d). Blue indicates the model transfers there; red indicates persistence wins. The colour bar is clipped at a robust bound so that one catastrophic failure does not compress the rest of the scale.

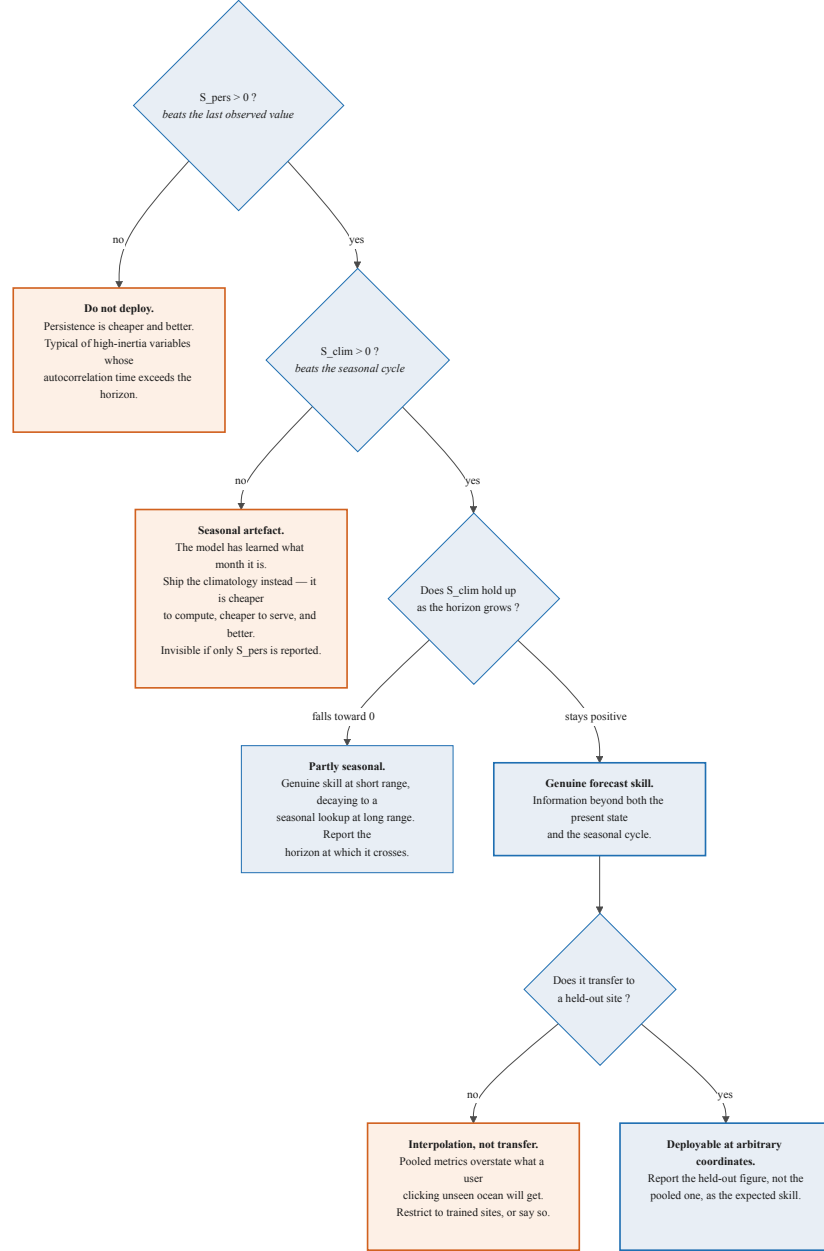


Fig.6. Interpreting the two skill scores together. Three of the six terminal states are negative recommendations that reporting S_{pers} alone would not reveal.